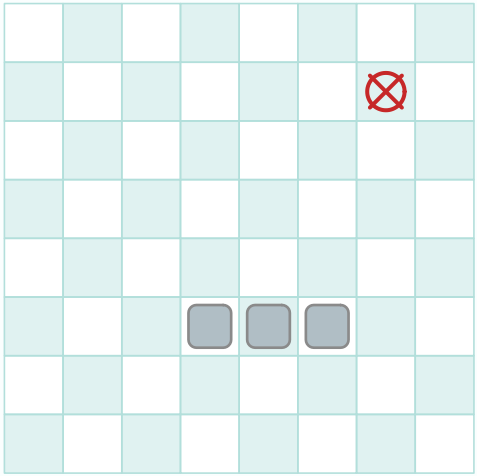
Chapter 19

Graphs of functions

a picture of every input and output at once



"along 7, up 7: hit!"
Every square has an address.
A graph is a grid where every
input gets a dot at
(input, output).

A **graph** turns a function into a picture. Each input x and its output $f(x)$ become one dot, and all the dots together draw a line or curve. You can then **see** things that are hard to spot in a formula: where it's biggest, where it crosses zero, whether it grows fast or slowly.

You'll start with the grid itself (Battleship), plot a curve by hand, learn to read a graph in both directions, and then meet the "family portraits" of the common functions from Lecture 7 so you recognise them on sight.

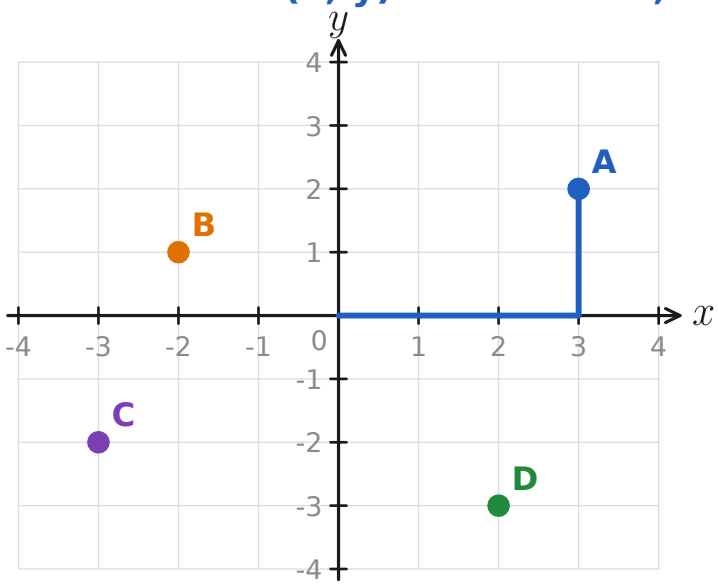
Goes with **Lecture 7** (graphing common functions, dominance), **Lecture 13** (graph of a function) and **Lecture 14** (intercepts and vertex).

Coordinates: Battleship with negative numbers

RULE: coordinates

The point (x, y) is x **across** (right if positive, left if negative) and then y **up** (down if negative), starting from the origin $(0, 0)$. Always x first.

Two number lines cross at right angles. The across one is the **x-axis**, the up one is the **y-axis**, and they meet at the **origin** (0, 0). A point's address is written **(x, y)**: **across first, then up**.



A = (3, 2): along 3, up 2 (blue path: corridor first, then the stairs)

B = (−2, 1): back 2, up 1

C = (−3, −2): back 3, down 2

D = (2, −3): along 2, down 3

Memory hook: "along the corridor, then up the stairs"

The axes split the page into four **quadrants**. Negative x means left of the y-axis; negative y means below the x-axis.

From table to dots to curve

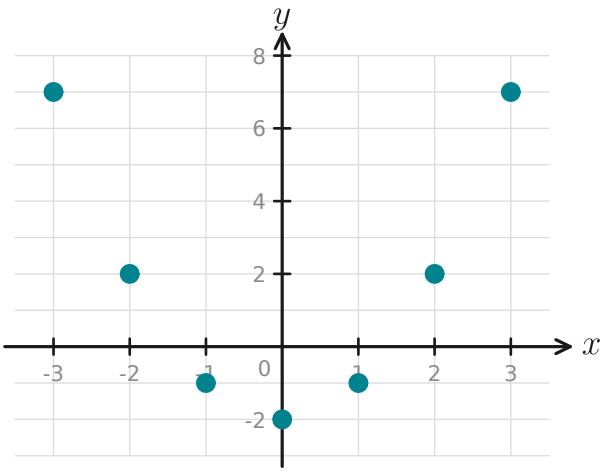
PROCEDURE: plotting y = f(x)

1. Choose inputs x (include negatives and 0) and work out each $y = f(x)$ in a table.
2. Plot each pair (x, y) as a dot.
3. Join the dots with a **smooth** curve (a ruler only if the rule is a straight line).

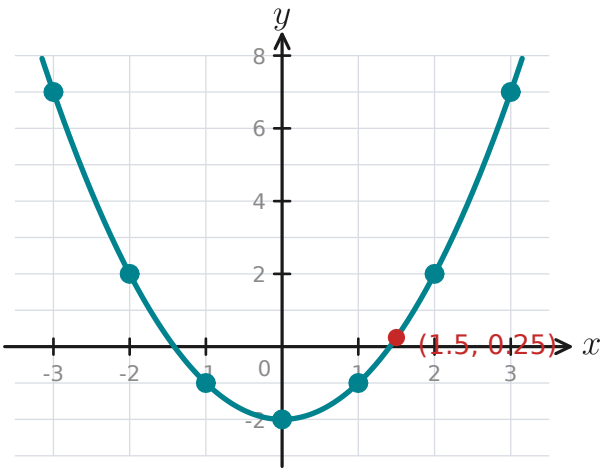
To draw $y = x^2 - 2$: choose some inputs, work out each output (a table), plot each pair as a dot, then join the dots with a smooth curve.

x	−3	−2	−1	0	1	2	3
y = x² − 2	7	2	−1	−2	−1	2	7

Careful with negatives: for $x = -3$, it's $(-3)^2 - 2 = 9 - 2 = 7$, not $-9 - 2$.



1. plot the dots



2. join with a smooth curve

Why smooth, not straight lines between dots? Because in-between inputs have outputs too: $x = 1.5$ gives 0.25, which lands right on the curve (red dot). Lecture 13 puts it neatly: the graph of f is **the set of all points** $(x, f(x))$.

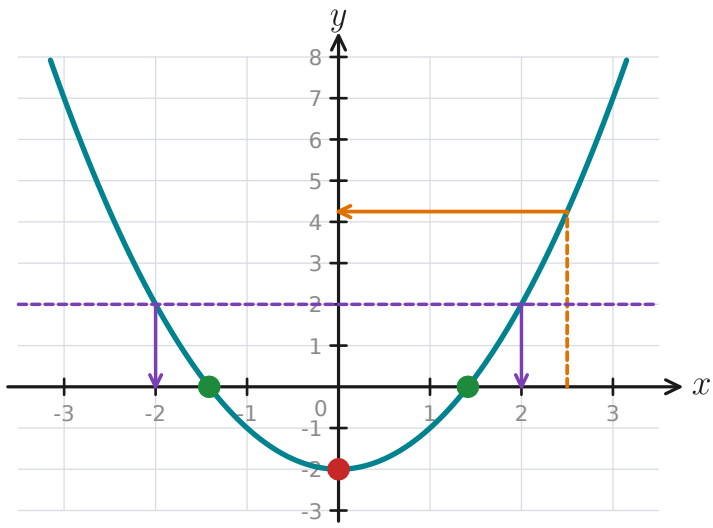
📈 Reading a graph both ways

RULE: intercepts and reading a graph

y-intercept: put $x = 0$; the graph crosses the y-axis at $(0, f(0))$.

x-intercepts (roots): put $y = 0$ and solve $f(x) = 0$.

To solve $f(x) = k$ from a graph: go across from k on the y-axis to the curve, then down to read x.



Input → output: from x = 2.5 go UP to the curve, then ACROSS: y ≈ 4.25

Output → input: which x give y = 2? go ACROSS from 2, then DOWN: x = −2 and x = 2

y-intercept (red): where it crosses the y-axis. Put x = 0: y = −2

x-intercepts (green): where y = 0, the solutions of $x^2 - 2 = 0$: x = ±1.41

That second direction is **solving an equation by picture**: the answers to $x^2 - 2 = 2$ are where the curve meets the horizontal line $y = 2$. (Chapter 15's crossing lines were the same idea.)

Even functions fold, odd functions spin

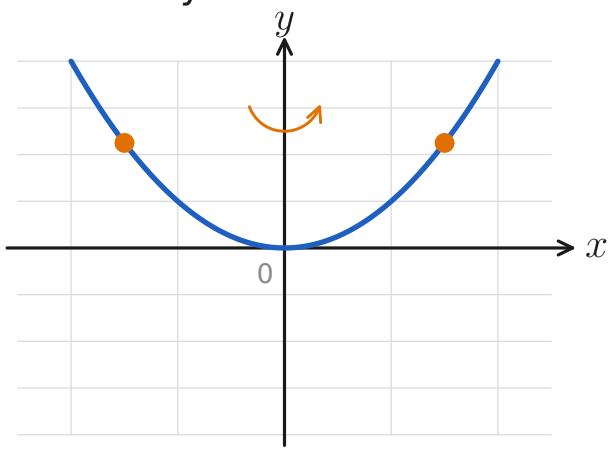
RULE: even and odd functions

Even: $f(-x) = f(x)$ for every x. The graph is symmetric in the y-axis (e.g. x^2, x^4, $x^2 + 3$).

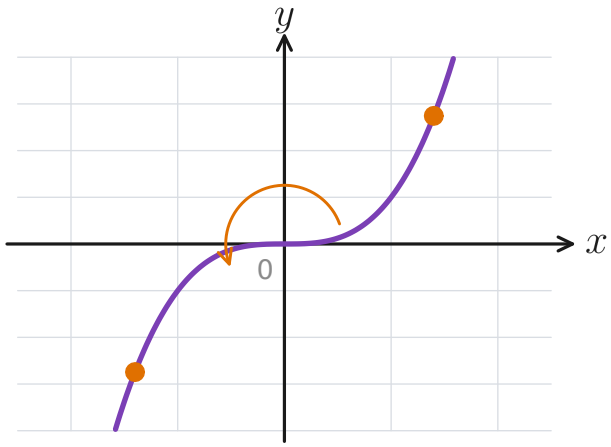
Odd: $f(-x) = -f(x)$ for every x. The graph is unchanged by a half turn about the origin (e.g. x, x^3, $\frac{1}{x}$).

To test: work out $f(-x)$ and simplify. Same as $f(x)$: even. Exactly $-f(x)$: odd. Otherwise: neither.

Lecture 7 sorts the power graphs into **even** and **odd**. You can test by picture or by numbers:



x²: fold along the y-axis



x³: spin half a turn about (0, 0)

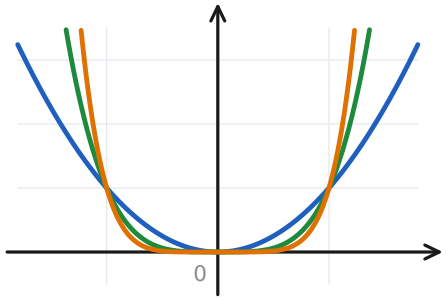
Even: fold the page along the y-axis and the two halves match. In numbers, putting in $-x$ gives the **same** output: $(-1.5)^2 = 1.5^2 = 2.25$. So $f(-x) = f(x)$.

Odd: spin the page half a turn around the origin and the curve lands on itself. In numbers, $-x$ gives the **negative** output: $(-2)^3 = -8$. So $f(-x) = -f(x)$.

The names come from the powers: x^2, x^4, x^6 are even functions; x, x^3, x^5 are odd. Many functions (like $x^2 + x$) are neither.

📈 Family portraits: recognise them on sight

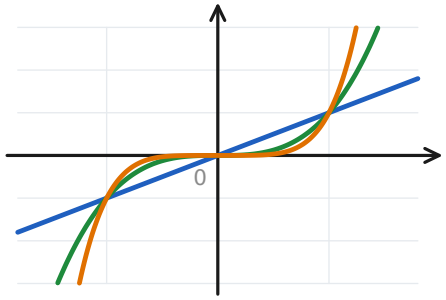
These are the shapes from Lecture 7. Learn them like faces; you'll meet them everywhere.



even powers

x^2 x^4 x^6

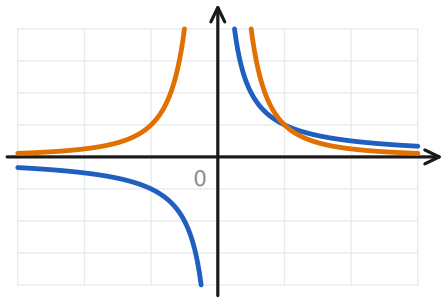
all pass through (−1, 1), (0, 0), (1, 1). Never negative. Higher power: flatter near 0, steeper beyond 1.



odd powers

x x^3 x^5

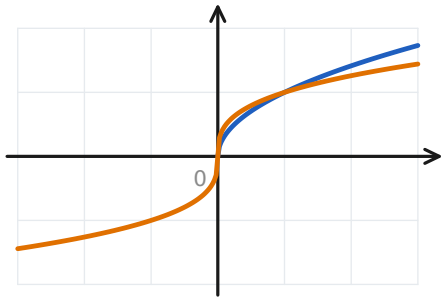
negative on the left, positive on the right; all pass through (−1, −1), (0, 0) and (1, 1).



negative powers

$1/x$ $1/x^2$

blow up near x = 0 (can't divide by 0), and creep towards 0 far out, never quite touching the axes.



fractional powers

$\sqrt{x}$ $\sqrt[3]{x}$

$\sqrt{x}$ only exists for $x \geq 0$ (half a curve); $\sqrt[3]{x}$ works for negatives too. Both grow slower and slower.

Those lines that $\frac{1}{x}$ creeps towards but never reaches are called **asymptotes**.

Dominance: what you see depends on the zoom

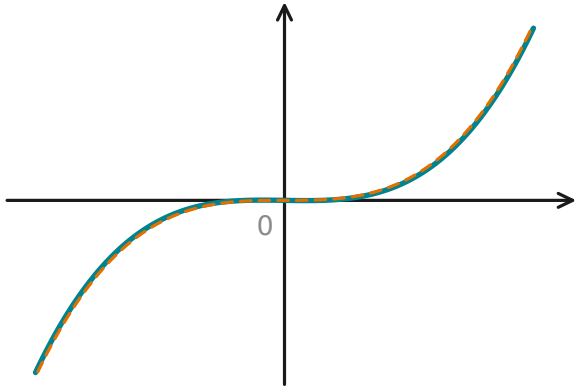
RULE: dominance (Lecture 7)

For $y = ax^m + \cdots + bx^n + c$ (highest power m, lowest power n):

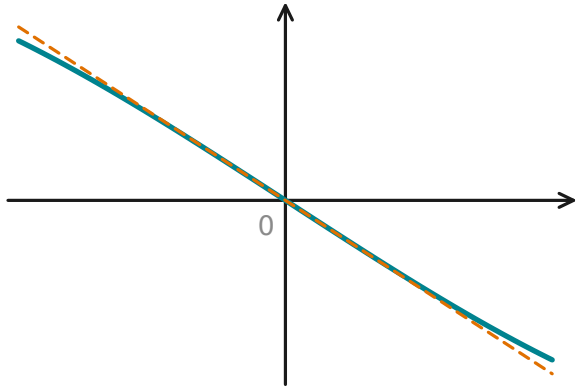
Large x (far from 0, either way): the **highest power** term dominates, so it looks like $y = ax^m$.

Small x (close to 0): the **lowest powers and the constant** dominate, so it looks like $y = bx^n + c$.

Lecture 7's example $y = x^3 - 2x$. Zoom out and the x^3 term swamps everything (the **biggest power dominates** for large x). Zoom in near 0 and x^3 is tiny, so it looks like the line $y = -2x$:



zoomed out: looks like x³ (dashed)



zoomed in: looks like −2x (dashed)

At $x = 10$: $x^3 = 1000$ and $2x = 20$, so the $-2x$ changes the answer by only 2%. At $x = 0.1$: $x^3 = 0.001$ and $2x = 0.2$, so now x^3 hardly matters.

Example: $y = x^4 - 3x + 2$. For large x it looks like $y = x^4$ (a U shape). Near 0 the x^4 is tiny and it looks like the straight line $y = -3x + 2$: at $x = 0.1$, $x^4 = 0.0001$ while $-3x + 2 = 1.7$.

📈 Sliding and flipping graphs

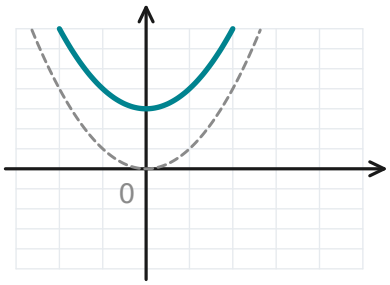
RULE: transformations of y = f(x)

A change **outside** the function (to the output, y) moves the graph **vertically** and does exactly what it says.

A change **inside** the brackets (to the input, x) moves the graph **horizontally** and does the **opposite** of what it looks like: $+a$ moves left, $\times a$ squashes.

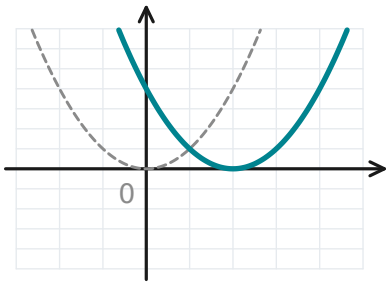
new graph	what happens to y = f(x)	point (x, y) moves to
$y = f(x) + a$	**up** a (down if $a < 0$)	$(x,\ y + a)$
$y = f(x + a)$	**left** a (opposite to the sign!)	$(x - a,\ y)$
$y = f(x - a)$	**right** a	$(x + a,\ y)$
$y = af(x)$	vertical stretch, factor a (from the x-axis)	$(x,\ ay)$
$y = f(ax)$	horizontal squash by factor a (a stretch of factor $\frac{1}{a}$, towards the y-axis)	$(\frac{x}{a},\ y)$
$y = -f(x)$	reflect in the x-**axis** (upside down)	$(x,\ -y)$
$y = f(-x)$	reflect in the y-**axis** (left-right)	$(-x,\ y)$

Change the formula a little and the whole picture moves. Starting from $y = x^2$:



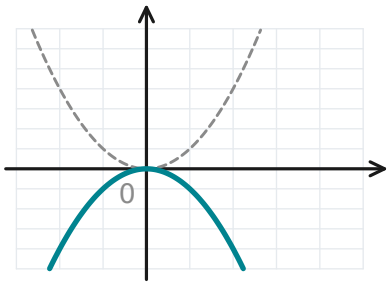
$y = x^2 + 3$

up 3: add 3 to every output



$y = (x - 2)^2$

RIGHT 2 (not left!): x must be 2 bigger to get the same output



$y = -x^2$

flipped upside down: every output changes sign

The one that feels backwards: $(x-2)^2$ moves **right**. Check with numbers: the lowest point of x^2 is at $x=0$; for $(x-2)^2$ the bracket is 0 when $x=2$. So the bottom has moved to $x=2$, to the right.

Chapter 15, explained: $y=x^2+6x+5=(x+3)^2-4$. That's x^2 moved **left 3** and **down 4**, so its lowest point (vertex) is at $(-3,-4)$, just as Lecture 14 said. The completed-square form is the **sat-nav** to the vertex.

RULE: vertex form of a quadratic

$y=a(x-h)^2+k$ has its **vertex** (turning point) at (h,k) and is symmetric about the line $x=h$.

$a>0$: U-shaped, the vertex is the **lowest** point. $a<0$: upside down, the vertex is the **highest** point.

Watch the sign of h: $(x+3)^2$ is $(x-(-3))^2$, so $h=-3$.

PROCEDURE: moving a point

1. Find the **horizontal** change from inside the bracket (opposite sign) and apply it to x.
2. Find the **vertical** change from outside and apply it to y (stretch first, then shift).
3. Check: put the new x into the new formula; it should give the new y.

Example: $(2,5)$ lies on $y=f(x)$. Where does it go on $y=f(x-3)+1$?

$x:\ 2\to 2+3=5$ ← inside: −3 means RIGHT 3

$y:\ 5\to 5+1=6$ ← outside: +1 means up 1

$(2,5)\ \to\ (5,6)$

Check with $f(x)=x^2+1$ (which does pass through (2, 5)): new formula at $x=5$ is $f(2)+1=6$ ✓.

Example: $(4,6)$ lies on $y=f(x)$. Where does it go on $y=f(2x)$, on $y=3f(x)$ and on $y=-f(x)$?

$f(2x):\ (4,6)\to(2,6)$ ← inside ×2 squashes: x ÷ 2

$3f(x):\ (4,6) \to (4,18)$ ← *outside ×3 stretches: y × 3*

$-f(x):\ (4,6) \to (4,-6)$ ← *flip in the x-axis*

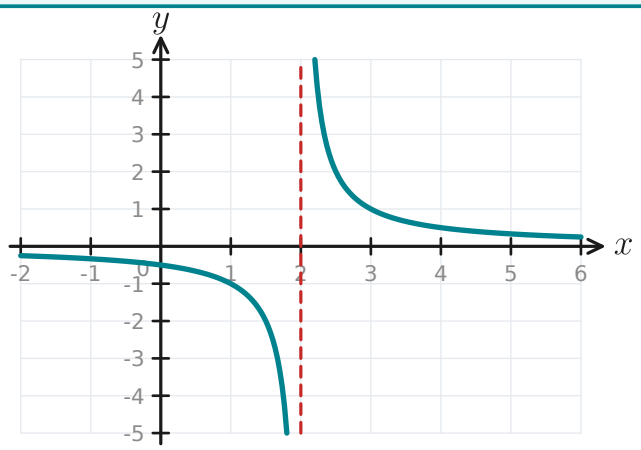
Asymptotes: where a graph can't go

RULE: asymptotes of fractions

A **vertical asymptote** is where the denominator is 0: $y = \dfrac{1}{x-a}$ has the asymptote $x = a$. The graph shoots up or down beside it and never touches it.

A **horizontal asymptote** is the value y approaches as x gets very large: for $\dfrac{1}{x-a}$ it is $y = 0$, because 1 divided by a huge number is nearly 0.

$y = \dfrac{b}{x-h} + k$ is $\frac{1}{x}$ moved: asymptotes $x = h$ and $y = k$.



y = 1/(x − 2)

red dashed: x = 2. Put x = 2 in and you'd divide by 0, so the graph can't be there.

Far left and far right it creeps towards the x-axis (y = 0).

Example: $y = \dfrac{3}{x+1} + 2$. Denominator 0 when $x = -1$: vertical asymptote $x = -1$. For large x the fraction is nearly 0, so y is nearly 2: horizontal asymptote $y = 2$.

Find the unknown: reading a formula from facts

PROCEDURE: partial-information graph problems

1. A point (p, q) on a graph means: put $x = p$ into the formula and you get $y = q$. That's an equation for the unknown.
2. A vertex (h, k) means: write $y = a(x - h)^2 + k$ and fill in h and k.
3. An asymptote $x = a$ means: the denominator is 0 when $x = a$.
4. A moved graph: use the transformation table, on the point or on the formula.

Example: $y = (x - h)^2 + k$ has vertex $(3, -2)$. Find h and k, and write the formula.

$$h = 3,\ k = -2: \quad y = (x - 3)^2 - 2$$ ← vertex (h, k) read straight off

Example: $y = a(x - 1)^2 + 2$ passes through $(3, 10)$. Find a.

$$10 = a(3 - 1)^2 + 2 = 4a + 2 \ \Rightarrow\ a = 2$$

put x = 3, y = 10

Example: the graph of $y = f(x)$ is moved to $y = f(x + 4)$. Where does the point $(6, 1)$ go?

$$(6, 1) \ \rightarrow\ (6 - 4,\ 1) = (2, 1)$$ ← inside +4: LEFT 4

Example: $y = x^2 + bx + 3$ is an even function. Find b.

$$f(-x) = x^2 - bx + 3 = x^2 + bx + 3 \ \Rightarrow\ -b = b \ \Rightarrow\ b = 0$$

even: f(−x) = f(x)

Example: $y = \dfrac{5}{x - k}$ has a vertical asymptote at $x = 4$. Find k.

$$4 - k = 0 \ \Rightarrow\ k = 4$$ ← denominator 0 at the asymptote

Mistakes to avoid

✗ Common mistakes

✗ $y = (x+2)^2$: right 2

inside the bracket: opposite way

✓ $y = (x+2)^2$: left 2

✗ $y = f(2x)$: $(4,6) \to (8,6)$

inside ×2 squashes: divide x by 2

✓ $(4,6) \to (2,6)$

✗ $y = (x+3)^2 - 4$: vertex $(3,-4)$

h is the number that makes the bracket 0

✓ vertex $(-3,-4)$

✗ $(-3)^2 - 2 = -9 - 2$

square the whole negative number

✓ $(-3)^2 - 2 = 9 - 2 = 7$

✗ point $(2,5)$: $y = 2,\ x = 5$

x first, then y

✓ $x = 2,\ y = 5$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Coordinates	$(x,\ y)$ across first, then up
Intercepts	y-intercept: put $x = 0$. x-intercepts: solve $f(x) = 0$
Even function	$f(-x) = f(x)$ symmetric in the y-axis
Odd function	$f(-x) = -f(x)$ half-turn symmetry about the origin
Dominance	large x: highest power wins. Near 0: lowest powers and the constant win
Asymptotes	$y = \dfrac{b}{x-h} + k: \quad x = h,\ y = k$ vertical asymptote where the denominator is 0
Vertex form	$y = a(x-h)^2 + k \ \Rightarrow\ \text{vertex}\ (h, k)$ $a > 0$ minimum, $a < 0$ maximum
Shift up / down	$y = f(x) + a$ up a: $(x, y) \to (x, y + a)$
Shift left / right	$y = f(x + a)$ **left** a: $(x, y) \to (x - a, y)$
Vertical stretch	$y = af(x)$ $(x, y) \to (x, ay)$
Horizontal squash	$y = f(ax)$ $(x, y) \to (\frac{x}{a}, y)$
Reflections	$y = -f(x), \quad y = f(-x)$ in the x-axis; in the y-axis

Battleship: your turn

Answers in grey. Sketch every graph on squared paper.

Points and plotting

1. Which quadrant is $(-4, 2)$ in? (top-right is the 1st, go anticlockwise)
2. Make a table for $y = 2x - 1$ with $x = -2$ to 2, and plot it.
3. For $y = x^2 - 4x$, find the y-intercept and the x-intercepts.

1. 2nd (top-left) 2. $y = -5, -3, -1, 1, 3$ (a straight line) 3. y-int 0; $x = 0$ and $x = 4$

Even, odd, neither

4. $y = x^4 + 1$
5. $y = x^3 - x$
6. $y = x^2 + x$
7. $y = 1/x$

4. even 5. odd 6. neither (try $x = 1$ and $x = -1$: 2 and 0) 7. odd

Recognise and move

8. Which family: $y = x^{-2}$?
9. Where is the vertex of $y = (x - 1)^2 + 5$?
10. Describe $y = (x + 4)^2$ compared with $y = x^2$.
11. Which term dominates $y = 5x^2 + x^4$ for large x?

8. negative powers (even: always positive) 9. (1, 5) 10. moved left 4 11. x^4

Find the unknown

12. $y = (x - h)^2 + k$ has vertex $(3, -2)$. Find h and k.
13. $(6, 1)$ is on $y = f(x)$. Where does it move to on $y = f(x + 4)$?
14. $(2, 5)$ is on $y = f(x)$. Where does it move to on $y = f(x + 1) - 4$?
15. $(4, -3)$ is on $y = f(x)$. Where does it move to on $y = -2f(x)$?
16. $y = a(x + 2)^2 - 1$ passes through $(0, 7)$. Find a.
17. $y = 3/(x - k)$ has a vertical asymptote at $x = -2$. Find k.

12. $h = 3$, $k = -2$ 13. (2, 1): inside +4 moves left 4 14. (1, 1): left 1, down 4 15. (4, 6): $y \times (-2)$ 16. $a = 2$ ($7 = 4a - 1$) 17. $k = -2$ (denominator $-2 - k = 0$)